Sébastien Barbier

Staff Software Engineer | Full-Stack: TypeScript, React, Python

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Staff-level software engineer with 16+ years shipping production systems, including agentic AI platforms. Led frontend for an agentic AI platform at Unique AG and SAP FSM's move to microfrontends. Currently building Shellui (TypeScript, React, Python/Django) and improving my German beyond B1.

Personal Project

Full-Stack Developer Jan 2026 – Oct 2026

Shellui (shellui.com): open-source microfrontend shell *Zürich, Switzerland*

- Built a React/TypeScript microfrontend shell and CLI that gives apps one navigation and theme.
- Unified sign-in across apps with OAuth, SAML SSO, magic links and Supabase Auth.
- Designed multi-tenant PostgreSQL schemas in Django services, isolating each tenant's data.
- Built signed webhooks and Celery jobs with retries and an outbox, so failed deliveries are retried.
- Protected every release with 1,000+ tests and CI/CD on each pull request (web and Tauri).

Professional Experience

Staff Frontend Engineer Dec 2023 – Dec 2025

Unique AG *Zürich, Switzerland / Remote*

- Led frontend for an agentic AI chat platform used by financial services clients.
- Streamed LLM answers live in the UI over GraphQL, with structured outputs and tool calling in production.
- Integrated RAG and MCP into the chat flows, so answers use client context and stay compliant.
- Partnered with product, design and backend teams on delivery, keeping release quality high.
- Used AI coding agents in the daily workflow to ship frontend changes with strong test coverage.

Liquidity Provider, Decentralized Finance (DeFi) Aug 2021 – Nov 2023

Private investor (own capital) *Zürich, Switzerland*

- Provided liquidity to on-chain DeFi AMM pools, fully funding my living costs for 2+ years.
- In parallel, maintained personal open-source apps in React and Django REST Framework.
- Built FromEdwin.com as a learning project on asynchronous performance metrics, owning architecture, backend, observability and CI/CD.

Associate Development Architect Feb 2020 – Jul 2021

SAP (Schweiz) AG, FSM; promoted from Technical Lead *Zürich, Switzerland / Remote*

- As Technical Lead, moved FSM to microfrontends, aligning several teams on one architecture.
- Co-designed the FSM extensibility concept for secure third-party integrations, driving partner adoption with code samples and public webinars.
- Prepared management decisions on FSM's target architecture by assessing risks and effort.
- Designed, coded and tested complex FSM features, and helped others resolve complex issues.
- Mentored junior developers and coached the team on architecture, simplicity and best practices.

Web Technologies Consultant and Freelancer Sep 2016 – Jan 2020

Self-employed *Chiang Mai, Thailand / Remote*

- Built the UI of Mozilla's Notes by Firefox extension and contributed to its React Native app.
- Delivered chatbots, PWAs and web integrations for clients including Chefclub and Mila AG.
- Worked remotely and independently with international clients, from first call to release.

Front-end Developer / Technical Lead Jun 2014 – Aug 2016
Fluance AG *Solothurn, Switzerland*

- Led frontend in a four-person Scrum team building a cross-platform HL7 patient data interface.
- Owned documentation, testing and deployment; stack: AngularJS, jQuery and Foundation CSS.

Earlier Experience

Java Developer Sep 2012 – Jun 2014
Covalia *Besançon, France*

- Maintained Java HL7/DICOM exchange for a telemedicine platform under ISO 13485.

Part-time Lecturer Sep 2011 and Sep 2012
University of Franche-Comté *Besançon, France*

- Taught 98 hours of practical Java and HTML/CSS to third-year engineering students.

Co-founder and Full-stack Developer Jul 2009 – Jun 2012
Ionyse *Besançon, France*

- Co-founded a web agency during my master's, building and hosting client websites on a Python/Django CMS.

Technical Skills

AI
Agentic AI, Autonomous systems, LLMs, streaming responses, structured outputs, tool calling, RAG and MCP.
AI-assisted engineering with AI coding agents as part of the daily workflow.
Frontend
TypeScript, JavaScript, React, Next.js, Angular, Tailwind CSS, PWA, accessibility, UX/UI.
Microfrontends, Redux, MUI, D3, streaming chat interfaces, React Native.
Backend & Architecture
Python, Django REST Framework, Java, REST APIs, GraphQL, OpenAPI, SSO, Celery, PostgreSQL, Supabase.
Multi-tenant SaaS, microfrontends, event-driven services, signed webhooks, retries, Redis.
Delivery & Ops
CI/CD, GitHub Actions, Docker, Cypress, unit and E2E testing, Sentry, Prometheus, Grafana.
Git, Linux server administration, Tauri desktop builds.
Domains
Financial services, healthcare (HL7, DICOM, ISO 13485), enterprise SaaS, open source.
Field service management (SAP), telemedicine (Covalia), decentralized finance (DeFi).

Soft Skills

Ownership: built Shellui end to end, from architecture and UI to CI/CD and hosting.

Collaboration: aligned several SAP teams on microfrontends; partnered with product and design at Unique.

Mentoring: mentored junior developers at SAP and taught 98 hours of programming at university.

Communication: presented the FSM extensibility concept to SAP partners in public webinars.

Pragmatism & quality: simple, well-tested solutions shipped in small steps, backed by 1,000+ tests.

AI delivery: shipped agentic AI chat at Unique with streaming LLMs, RAG, and MCP for compliant financial-services clients.

Education

Master's in Computer Science 2011
Distributed Systems and Networks, University of Franche-Comté, France

Bachelor's in Computer Science 2009
University of Franche-Comté, France

Languages

French: Native | English: Fluent | German: B1 (Goethe-Zertifikat, July 2026)